

USER MANUAL

AUTOMATIC TOUCHLESS WATER DISPENSER

Phase 1 Prototype — Setup, Operation, Safety & Troubleshooting

Controller: Arduino Uno

Detection: HC-SR04 Ultrasonic Sensor

Interface: 16×2 I2C LCD

Actuation: Dual Servo Mechanism

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1. Important Notice

This manual describes the current Phase 1 prototype based on the supplied Arduino program and project design. Operate it only after checking wiring, power, servo linkage and the water path. This prototype is not a certified commercial appliance. Keep exposed electronics and electrical connections away from water.

2. Product Overview

The Automatic Touchless Water Dispenser operates a tap mechanism without requiring direct contact. An HC-SR04 ultrasonic sensor measures distance. When a valid hand/object is detected within approximately 15 cm, the Arduino Uno commands two servo motors to move the mechanism to its open position. The LCD displays system status. After approximately three seconds, the servos return to the closed position.

3. Main Components

Component	Function
Arduino Uno	Controls sensing, display logic and servo operation.
HC-SR04	Detects a hand/object by measuring distance.
16×2 I2C LCD	Displays instructions, distance and status.
Two servo motors	Operate the tap mechanism.
Power source	Supplies the prototype according to the final power design.
3D enclosure	Supports/protects electronics and mechanical parts.

4. Before First Use

Confirm that all wiring matches the project circuit. Check that the servo linkage moves freely and that 0° corresponds to the mechanically closed position. Keep exposed electronics away from possible splashes. Use only the intended power arrangement for the assembled prototype.

5. Operating Instructions

Step	Instruction
1	Power on the assembled dispenser using its intended power source.
2	Wait for the LCD startup message: “Automatic Water / Dispenser”.
3	The LCD then displays “Place Hand” and the measured distance.
4	Place a hand in front of the ultrasonic sensor within approximately 15 cm without touching the sensor.
5	When detection is valid, the LCD shows “Dispensing / Water...” and both servos move from 0° toward 90°.
6	The programmed dispensing interval lasts approximately three seconds.
7	The servos return to 0°. The LCD shows “Thank You / Visit Again”.
8	Wait until the mechanism is fully closed before servicing or moving the unit.

6. LCD Messages

Message	Meaning
Automatic Water / Dispenser	Startup.
Place Hand	Ready for detection.
Dist: xx.x cm	Measured ultrasonic distance.
Dispensing / Water...	Servo dispensing sequence is active.
Thank You / Visit Again	Sequence completed and servos commanded closed.

7. Safety Precautions

Keep electronics, connectors and power components dry. Disconnect power before changing wiring, adjusting the servo linkage or opening the enclosure. Do not force powered servos by hand. Keep fingers away from moving linkages. Do not use damaged batteries, wires or connectors. Use only an appropriate supply for the actual assembled components. For drinking-water use, every part directly contacting water must be suitable for that purpose and maintained hygienically.

8. Basic Maintenance

Disconnect power first. Keep the ultrasonic sensor face clean and unobstructed. Inspect servo linkages for looseness, binding or misalignment. Check wiring and connectors periodically. Keep the enclosure dry. Do not spray cleaning liquid directly onto electronics.

9. Troubleshooting

Problem	Check
LCD not working	Check power, ground, I2C wiring and configured address 0x27.
Distance incorrect	Check HC-SR04 power, TRIG D9, ECHO D10, mounting and obstructions.
Servos not moving	Check D6/D7 signals, servo power, common ground and mechanical jamming.
Tap movement incorrect	Verify linkage and that 0° is closed and 90° is the intended open position.
Unexpected dispensing	Remove objects within 15 cm and check sensor reflections/mounting.
System resets with servo motion	Check whether the power source can supply servo current and inspect voltage drop/wiring.
Water does not stop	Safely isolate power/water and inspect the mechanical valve/linkage; do not rely on software if the me

10. Technical Parameters

Parameter	Phase 1 setting
Detection threshold	15 cm
Servo closed / open	0° / 90°
Dispensing interval	3 seconds
HC-SR04 TRIG / ECHO	D9 / D10
Servo signals	D6 / D7
LCD	16×2 I2C, address 0x27
LCD SDA / SCL	A4 / A5 on Arduino Uno

11. Shutdown and Storage

Ensure the mechanism is closed, then switch off or disconnect the intended power source. Store the prototype in a dry location. Do not store it with wet electronics or damaged power components.

12. Phase 1 Limitations

The current system uses a fixed 15 cm threshold and a fixed three-second interval. It does not currently confirm cup presence, measure water level or flow, detect leaks, or verify mechanically that the tap closed. These functions should not be assumed unless added and tested.

13. Related Documentation

Refer to the Phase 1 Project Report and project website for circuit information, CAD design, source code and development details. The website can provide this manual, the technical report, Arduino source and STL models.